The operator-algebra arc of the GeoVac framework: a reading guide for spectral triples, Latrémolière propinquity, and Lorentzian convergence

J. Loutey*

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Abstract

We provide a unified reading of the eleven papers that make up the operator-algebra arc of the GeoVac framework (Papers 29, 32, 38, 39, 40, 42, 43, 44, 45, 46, 47). The arc constructs an explicit almost-commutative spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected unit three-sphere with a Camporesi–Higuchi spinor Dirac operator, places it in the Marcolli–van Suijlekom gauge-network lineage, and proves four load-bearing theorems. First, the underlying truncated graphs are Ramanujan, with integer-algebraic Ihara zeros (Paper 29). Second, the Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}}$ converge to the continuum spectral triple $\mathcal{T}_{S^3}$ in the Latrémolière propinquity at quantitative rate $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \le C_3\gamma_{n_{\max}}$ with $C_3 = 1$ and asymptotic constant $4/\pi$ (Paper 38). Third, the rate constant $4/\pi$ is universal across the class of simple compact Lie groups with bi-invariant metric, via a Plancherel-weight $\times$ Vandermonde-Jacobian cancellation in the dual-Coxeter normalisation (Paper 40). Fourth, a K^+-restricted weak-form Lorentzian propinquity convergence theorem holds on truncated $\mathrm{SU}(2) \times \mathrm{U}(1)_t$ Krein spectral triples (Paper 45) – to the author's knowledge the first such convergence result in the math.OA literature, closing a question deferred by Connes and van Suijlekom [7]. The convergence has since been strengthened in two directions: Paper 46 closes the strong-form Lorentzian propinquity convergence (without K^+ restriction) on the natural and enlarged operator-system substrates, and Paper 47 closes the de-compactification limit $T \to \infty$ at the norm-resolvent / spectral level via a two-rate hybrid composite. A master theorem (§9) subsumes Papers 38, 39, and 40 as special cases: for arbitrary $k \ge 1$ and arbitrary compact connected Lie groups $G_1, \ldots, G_k$, the k-fold tensor product propinquity converges at the universal $4/\pi$ rate with k-INDEPENDENT Lipschitz comparison constant. The constant $4/\pi$ identifies, in both Riemannian (Paper 38) and Lorentzian (Paper 43 §10.2 / Paper 46) settings, as the M_1 Hopf-base measure signature of the master Mellin engine catalogued by the case-exhaustion theorem of Paper 32 § VIII. This synthesis gathers the statements, sketches the load-bearing proofs, and exhibits the structural relationships between the papers; no new theorems are introduced.

1 Introduction

The GeoVac framework is a discretization scheme for separable quantum systems on natural geometries. When the geometry is the round $S^3 = \mathrm{SU}(2)$, the framework produces a sequence of finite truncations whose continuum limit is the Camporesi–Higuchi spectral triple of the round three-sphere [4]. The present paper synthesizes the nine documents in the project's operator-algebra group: Papers [32–40]. Together they form a self-contained research thread in mathematical operator algebras with four headline results.

*Independent researcher. jloutey@gmail.com. This synthesis was prepared in an AI-augmented agentic workflow with Anthropic's Claude. All design decisions and editorial choices are the author's responsibility.

Headline 1: the Hopf graphs are Ramanujan. The finite Fock-projected S^3 Coulomb graphs and the Bargmann–Segal S^5 holomorphic graphs satisfy the Kotani–Sunada Ramanujan bound at every tested cutoff. Their Ihara zeta zeros are algebraic integers with integer coefficients; no π or transcendental content appears in the Ihara polynomial factorizations (Paper 29 [32], Theorems 5.1–5.2 and Corollary 5.3).

Headline 2: Riemannian propinquity convergence with explicit rate $4/\pi$. The Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ of the SU(2) Camporesi–Higuchi spectral triple converge to the continuum triple $\mathcal{T}_{S^3}$ in the Latrémolière quantum Gromov–Hausdorff propinquity [15, 16], with

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; C_3 \cdot \gamma_{n_{\max}}, \qquad C_3 = 1, \qquad \gamma_{n_{\max}} \sim \frac{4}{\pi}\,\frac{\log n_{\max}}{n_{\max}} \quad (n_{\max} \to \infty). \tag{1}$$

The asymptotic constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ identifies as the Hopf-base measure of $S^3 \to S^2 \times S^1$ and is discussed in detail in §4–§8 below (Paper 38 [34], Theorems 3.4 and 4.1).

Headline 3: the rate constant $4/\pi$ is universal across compact simple Lie groups. For any simple compact Lie group G with bi-invariant metric (so $G \in \{\mathrm{SU}(N), \mathrm{Spin}(n), \mathrm{Sp}(n), G_2, F_4, E_6, E_7, E_8\}$), the Plancherel-weight $\times$ Vandermonde-Jacobian cancellation theorem (Paper 40 §3.2 [36]) leaves exactly the rank-1 Stein–Weiss constant $4/\pi$ from each positive root direction's Euler–Catalan asymptote, in the dual-Coxeter normalisation. The constant $C_3 = 1$ remains asymptotically tight at all ranks via the PRV-summand bound of Kumar (1988) and Vinberg (1990), applied to the dominant image of the tensor product $V_\lambda \otimes V^*_{\lambda'}$.

Headline 4: first Lorentzian propinquity convergence theorem. On truncated $\mathrm{SU}(2) \times \mathrm{U}(1)_T$ Krein spectral triples $\mathcal{T}^L_{n_{\max},N_t,T}$ built via the Bizi–Brouder–Besnard [2] signature $(3,1)$ data at $(m,n) = (4,6)$, the K^+-restricted weak-form Lorentzian propinquity satisfies

$$\Lambda^L(\mathcal{T}^L_{n_{\max},N_t,T}, \mathcal{T}^L_{\mathcal{M}}) \;\leq\; C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \;\longrightarrow\; 0, \tag{2}$$

with $\gamma^{\mathrm{joint}}_{n_{\max},N_t,T} = O(\log n_{\max}/n_{\max} + T/N_t)$, where $\Lambda^L(\cdot,\cdot) := \Lambda(\cdot|_{\mathcal{K}^+}, \cdot|_{\mathcal{K}^+})$ is the propinquity of the standard metric spectral triple obtained from the Krein-positive subspace $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$. This is, to the author's knowledge, the first Lorentzian propinquity convergence theorem in the math.OA literature (Paper 45 [40], Theorem 5.1). Connes and van Suijlekom [7] deferred this question three times to "elsewhere"; the present arc closes it at the K$^+$-weak-form level on truncated Krein spectral triples.

Position against the literature. The arc sits in the lineage of Connes–van Suijlekom 2021 [7], which formalised spectral truncations as operator systems with finite-cutoff Connes distance; of Marcolli and van Suijlekom 2014 [18], which introduced gauge networks as a discrete-graph realisation of finite spectral triples carrying connection on edges; and of the subsequent Latrémolière–Hekkelman–McDonald–Toyota–Farsi–Leimbach school of quantitative propinquity convergence for spectral truncations [8, 9, 12–15, 17, 28]. All these earlier convergence results are strictly Riemannian (and almost all are on flat structures S^1, $\mathbb{T}^d$, S^2 Berezin–Toeplitz). The Paper 38 SU(2) result and the Paper 40 generalisation are the first non-abelian non-flat instances at quantitative rate; the Paper 45 result is the first Lorentzian instance. Mondino and Sämann's synthetic Lorentzian Gromov–Hausdorff program [19, 20] is categorically distinct in object class (pre-length spaces with causal diamonds, not operator-algebraic spectral triples) and is not in competition with the present arc.

What this synthesis provides. A working mathematician reading the corpus cold may find the nine-paper sequence difficult to navigate. This document collects the constructions, restates the load-bearing theorems with cross-references back to the original papers, and exhibits the relationships between them. No new theorems are introduced. The reader is assumed familiar with spectral triples [5, 6], Lipschitz seminorms and Rieffel quantum metric spaces, and the Latrémolière propinquity [15, 16]. Familiarity with [7] is helpful but not strictly required. No knowledge of the GeoVac physics-side corpus (quantum chemistry, QED, precision spectroscopy) is assumed; this arc is self-contained from the operator-algebra side.

2 The GeoVac spectral triple

2.1 The Camporesi–Higuchi data on S^3

We work throughout on the unit three-sphere S^3, identified with SU(2) under its bi-invariant round metric. Following Camporesi and Higuchi [4], the spinor bundle is parallelisable, of rank 2, and the Dirac operator D_{CH} has spectrum

$$\mathrm{spec}(D_{\mathrm{CH}}) = \{\pm(n + \tfrac{3}{2}) : n \in \mathbb{N}_0\}, \qquad g_n^{\mathrm{Dirac}} = 2(n+1)(n+2) \tag{3}$$

on each chirality of the half-integer Hurwitz-shifted shell. We write $\mathcal{H}_{S^3} = L^2(S^3, \Sigma)$ for the chirality- doubled spinor Hilbert space carrying the Camporesi–Higuchi data.

2.2 The GeoVac algebra, Hilbert space, and Dirac operator

The GeoVac construction (Paper 32 §3 [33]) takes the following data:

- $\mathcal{A}_{\mathrm{GV}} \subset C^\infty(S^3)$ is the involutive algebra of complex-valued smooth functions on S^3, with the standard pointwise product and complex conjugation;
- $\mathcal{H}_{\mathrm{GV}} = \mathcal{H}_{S^3}$, with $\mathcal{A}_{\mathrm{GV}}$ acting by pointwise multiplication;
- $D_{\mathrm{GV}} = D_{\mathrm{CH}}$, the Camporesi–Higuchi Dirac operator;
- J_{GV} is the charge-conjugation real structure inherited from the spin structure of S^3, satisfying $J_{\mathrm{GV}}^2 = -I$ (Euclidean KO-dimension 3), $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$.

The triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}}; J_{\mathrm{GV}})$ is real, odd (no $\mathbb{Z}_2$-grading; KO-dimension 3), and has compact resolvent $|D_{\mathrm{GV}}|^{-1}$ with eigenvalue multiplicities given by (3). The Connes axiom audit is recorded in Paper 32 §4 [33], where each of the seven axioms (order zero, bounded commutators, compact resolvent, order one, real structure, $\mathbb{Z}_2$-grading by convention, and finite summability) is checked.

2.3 The Marcolli–van Suijlekom lineage

The role of GeoVac in the ecosystem of finite-and-graph spectral triples is explained in Paper 32 §7. The Marcolli–van Suijlekom construction [18] attaches a finite spectral triple to each vertex of a graph and a connection (gauge group element) to each edge; the spectral action of the resulting object recovers Wilson lattice gauge theory. The continuum-limit issue addressed by Perez-Sanchez [22, 23] clarifies that the limit is Yang–Mills without Higgs by default; a Higgs sector appears only via off-diagonal data on the finite factor.

The GeoVac Fock-projected S^3 graph fits this template naturally. Its vertices are the Fock index (n, ℓ, m) with $n = 2j + 1$, ℓ the orbital, and m the magnetic quantum number, providing a Peter–Weyl-indexed basis on $S^3 = \mathrm{SU}(2)$. Each Fock shell carries a finite spectral triple (roughly, a finite-dimensional subrepresentation of the Camporesi–Higuchi triple), and the edge structure encodes the SO(4) selection rules of the underlying graph. The graph admits a $U(1)$

Wilson construction (recorded as GeoVac internal Paper 25), an SU(2) Wilson construction on $S^3 = \mathrm{SU}(2)$ (Paper 30), and an SU(3) Wilson construction on the Bargmann–Segal S^5 [31], all of which carry the universal kinetic coefficient $1/(4N_c)$.

The Connes axiom audit (Paper 32 §4 [33]) checks explicitly that $J^2 = -I$ and $JD = +DJ$ hold on the truthful Camporesi–Higuchi data at every finite cutoff $n_{\max} \in \{1, 2, 3\}$ tested. These two relations, together with the parallelisability of $S^3 = \mathrm{SU}(2)$, distinguish the GeoVac triple from the Connes–van Suijlekom circulant comparator, which is commutative and gives propagation number 1.

2.4 Truncations as operator systems

For each $n_{\max} \in \mathbb{N}$, the orthogonal projection $P_{n_{\max}} : \mathcal{H}_{\mathrm{GV}} \to \mathcal{H}_{\mathrm{GV}}$ onto the span of Peter–Weyl modes with $n \le n_{\max}$ defines a truncated operator system

$$\mathcal{O}_{n_{\max}} := P_{n_{\max}} \mathcal{A}_{\mathrm{GV}} P_{n_{\max}} \subset \mathcal{B}(P_{n_{\max}} \mathcal{H}_{\mathrm{GV}}), \tag{4}$$

which is self-adjoint, contains the identity, but is in general *not* multiplicatively closed. The defect is precisely the content of the Connes–van Suijlekom operator-system formalism [7]: in the natural envelope, $\mathcal{O}_{n_{\max}}$ has propagation number $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ at every tested $n_{\max} \in \{2, 3, 4\}$, matching the Toeplitz S^1 result [7, Proposition 4.2] verbatim. A concrete witness pair at $n_{\max} = 2$ is $M^2_{2,1,0} \notin \mathcal{O}_2$ with 14.9% Frobenius residual against $\mathcal{O}_2$, deficit concentrated on the top-shell diagonal in the natural Avery basis (Paper 32 §3.4 [33]). The truncated triple is then

$$\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, P_{n_{\max}} \mathcal{H}_{\mathrm{GV}}, P_{n_{\max}} D_{\mathrm{GV}} P_{n_{\max}}),$$

with continuum limit $\mathcal{T}_{S^3} = (\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$.

3 The Hopf graphs as Ramanujan substrate

Before turning to convergence, we record the most basic spectral-graph property of the underlying finite truncations.

Ihara zeta of the Fock graph. For each $n_{\max}$, let $G_{n_{\max}}$ denote the directed Fock-projected S^3 Coulomb graph: vertices are triples (n, ℓ, m) with $n \le n_{\max}$, $\ell < n$, $|m| \le \ell$; edges encode SO(4) selection rules. The Ihara zeta function $\zeta_{G_{n_{\max}}}(s)$ is computed in Paper 29 §4 [32] via the Ihara–Bass determinantal formula and via the $2E \times 2E$ Hashimoto non-backtracking edge operator.

Theorem 3.1 (Paper 29, Theorems 5.1–5.2)**.** *For every $n_{\max} \in \{2, 3, 4, 5\}$, the Fock-projected S^3 graph $G_{n_{\max}}$ satisfies the Kotani–Sunada Ramanujan bound strictly: the largest non-trivial Hashimoto eigenvalue sits strictly inside the spectral radius $\sqrt{q_{\max}}$ associated with the maximum (Perron) vertex degree. All non-trivial Ihara zeros are algebraic integers; the corresponding Ihara polynomials factor over $\mathbb{Z}[s]$ into integer-coefficient pieces, and several factorizations admit explicit closed forms (e.g. $(4s^2 + 1)^2$, $(9s^2 + 1)^4$, the $12 + 22$ dichotomy for S^5 at $N_{\max} = 3$).*

Structural consequences. Theorem 3.1 secures two facts about the truncated graphs that are used implicitly elsewhere in the arc. First, no π or transcendental quantity appears anywhere in the Ihara polynomials, in any of the tested factorizations (Paper 29 §5.4 [32], Corollary on integer-algebraicity). This is the graph-spectral analog of the π-source case-exhaustion theorem (Paper 32 §8): both say that π enters only through specific named mechanisms (Hopf-base measure, spectral action, vertex-parity Hurwitz character), never in the bare combinatorial data. Second, the Hashimoto-spectrum Poisson statistics (Paper 29 §6.3) are categorically distinct from the GUE-like statistics seen on the spectral-zeta side of the framework, making clear that the Weil dictionary between Selberg and Riemann breaks at GeoVac by design.

The Ramanujan property of the truncated graphs underwrites our freedom to take naive sums and traces over $G_{n_{\max}}$ without worrying about pathological spectral gaps: the cutoff graphs are well-conditioned for spectral analysis, and the Ihara polynomials are entirely in $\mathbb{Z}[s]$. This is exactly the kind of substrate one wants under any convergence theorem to follow.

4 Riemannian Gromov–Hausdorff convergence: the central theorem

4.1 Statement (Paper 38)

The load-bearing theorem of the arc is Paper 38 [34], Theorem 3.4.

Theorem 4.1 (Paper 38, GH convergence on $S^3 = \mathrm{SU}(2)$)**.** *The Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}}$ of the round* $\mathrm{SU}(2)$ *Camporesi–Higuchi spectral triple converge to $\mathcal{T}_{S^3}$ in the Latrémolière quantum Gromov–Hausdorff propinquity:*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; C_3 \cdot \gamma_{n_{\max}} \;\to\; 0 \quad (n_{\max} \to \infty), \tag{5}$$

with $C_3 = 1$ (asymptotic tight via the PRV summand bound) and asymptotic constant $4/\pi$:

$$n\, \gamma_n \;\sim\; (4/\pi)\, \log n \qquad (n \to \infty). \tag{6}$$

A uniform finite-cutoff bound $\gamma_n \leq 6\,(\log n)/n$ holds for all $n \geq 2$.

The proof proceeds via five lemmas L1′, L2, L3, L4, L5 of Paper 38 §4–§6. We sketch each.

4.2 Lemma L1′: chirality-doubled truncated operator system

The original Connes–van Suijlekom truncation is on the scalar Fock graph; the spinor lift to the full Camporesi–Higuchi bundle (with both chiralities) is the subject of an internal sprint (R3.5). The truthful Camporesi–Higuchi data at finite $n_{\max}$ has the property that on most cross-shell pairs the Connes distance is $+\infty$ (an artefact of the n-shell degeneracy in the Avery basis); this is repaired by lifting to the full Dirac sector with both chiralities and off-diagonal shell-coupling multipliers. The chirality-doubled operator system has dimension $\dim \mathcal{O}_{n_{\max}} = 1, 14, 55, 140, \ldots$ at $n_{\max} = 1, 2, 3, 4, \ldots$ and every cross-shell pair has finite Connes distance. See Paper 38 §4.1 [34] for the explicit construction.

4.3 Lemma L2: central spectral Fejér kernel on $\mathrm{SU}(2)$

The propinquity tunneling pair built in L5 below requires a positive, state-preserving, central good kernel on SU(2) whose mass-concentration moment vanishes at a known rate. Paper 38 §4.2 constructs the central spectral Fejér kernel

$$K_{n_{\max}}(g) \;=\; \frac{1}{Z_{n_{\max}}} \left| \sum_{j \leq j_{\max}} \sqrt{2j+1}\, \chi_j(g) \right|^2, \qquad j_{\max} = (n_{\max}{-}1)/2, \quad Z_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2}. \tag{7}$$

The Plancherel symbol on the Avery–Wen–Avery basis is $\widehat{K}(j) = (2j+1)/Z_{n_{\max}}$ for $j \leq j_{\max}$. The cb-norm of the associated Herz–Schur multiplier acting on the group C^*-algebra is

$$\|S_{K_{n_{\max}}}\|_{\mathrm{cb}} \;=\; \frac{2}{n_{\max}+1} \;=\; O\big(1/n_{\max}\big), \tag{8}$$

via the Bożejko–Fendler central-multiplier equality [3] on amenable compact groups (Pier [24]; Paulsen [21]; Pisier [25]).

The asymptotic constant $4/\pi$. The mass-concentration moment $\gamma_{n_{\max}}$ of the kernel admits a closed-form decomposition via Stein–Weiss-type integration by parts [27], leading to the *Stein–Weiss sharpening* (Paper 38, Appendix A):

$$\gamma_{n_{\max}} \; = \; \pi - \frac{4\,T_{n_{\max}}}{\pi\,Z_{n_{\max}}}, \quad \text{with } \frac{n\,\gamma_n}{\log n} \to \frac{4}{\pi} \text{ as } n \to \infty. \tag{9}$$

The constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ is the *Hopf-base measure factor* in the SU(2)/U(1) Haar normalisation. Concretely, for the Hopf bundle $S^3 \to S^2$, $\mathrm{Vol}(S^2) = 4\pi$ and the right-hand side of (9) normalises by π^2 from the SU(2) character integration $\int_0^{2\pi} \sin^2(\theta/2)\,d\theta$. The expected one-log slowdown of the non-abelian SU(2) rate compared with the Leimbach–van Suijlekom torus rate $O(1/n_{\max})$ [17] is precisely the $\log n$ factor in (9).

4.4 Lemma L3: Lipschitz comparison with $C_3 = 1$

For each $f \in C^\infty(S^3)$ and each truncated multiplier M_f on $\mathcal{O}_{n_{\max}}$, the Lipschitz comparison

$$\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \;\le\; C_3 \cdot \|\nabla f\|_{L^\infty(S^3)} \tag{10}$$

holds with $C_3 = 1$ (Paper 38, Theorem 5.2). The proof uses SO(4) selection rules: the off-diagonal element of $[D_{\mathrm{CH}}, M_f]$ between Peter–Weyl shells V_λ and $V_{\lambda'}$ is bounded by $\sqrt{C(\lambda)} - \sqrt{C(\lambda')}$ where $C(\lambda)$ is the quadratic Casimir of the highest weight; the Dirac-triangle inequality $|D(\pi) - D(\pi')| \le \sqrt{C(\sigma)}$ on intertwining summands $\sigma \subset \pi \otimes \pi'^*$ secures $C_3 \le 1$, and the Avery harmonic gradient scaling forces $C_3 = 1$ in the limit $n_{\max} \to \infty$. At any finite $n_{\max}$, $C_3 = (N-1)/\sqrt{N^2 - 1}$ for $N = n_{\max}$, which converges to 1 from below.

The same bound is proved for all simple compact Lie groups in Paper 40 §3.3 [36], with asymptotic tightness $C_3(G) = 1$ at all ranks; see §5 below.

4.5 Lemma L4: Berezin reconstruction

A Berezin map $B_{n_{\max}} : C(S^3) \to \mathcal{O}_{n_{\max}}$ is constructed in Paper 38 §5.3 as

$$B_{n_{\max}}(f) \;=\; \sum_{N \le n_{\max},\, L,\, M} \widehat{K}(N)\, c_{NLM}\, M_{NLM}, \quad \widehat{K}(N) = N/Z_{n_{\max}}. \tag{11}$$

This is the SU(2) non-Kähler analog of the Connes–van Suijlekom [7] Fejér kernel reconstruction on S^1 and the non-holomorphic analog of Hawkins [11]. Four properties are established:

(a) Positivity: for $f \ge 0$, $B_{n_{\max}}(f)$ is a positive operator on $P_{n_{\max}}\mathcal{H}_{\mathrm{GV}}$ (via convolution + positivity-preserving compression);

(b) Contractivity: $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \le \|f\|_\infty$;

(c) Approximate identity: $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \le \gamma_{n_{\max}} \|\nabla f\|_\infty$ with $\gamma_{n_{\max}}$ as in L2;

(d) L3 compatibility: $\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \le \|\nabla f\|_\infty$ (i.e. $C_3 = 1$ is inherited via $\widehat{K} \le 1$).

This is the operator-system analog of the Berezin–Toeplitz quantisation; see Hekkelman–McDonald [13] for the closely related $\mathbb{T}^d$ case.

4.6 Lemma L5: Latrémolière propinquity assembly

The tunneling pair $(B_{n_{\max}}, P_{n_{\max}})$ between $\mathcal{T}_{S^3}$ and $\mathcal{T}_{n_{\max}}$ has reach $\le \gamma_{n_{\max}}$ (by L4(c) approximate identity and L2(g) Bożejko–Fendler dual estimate), and height contributions are bounded by L4(b) contractivity and the fact that the projection P contributes $\mathrm{height}_P = 0$. The assembly machinery is Latrémolière [15], applied verbatim. This gives

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; C_3 \cdot \gamma_{n_{\max}} \;\to\; 0, \tag{12}$$

completing the proof of Theorem 4.1.

4.7 Numerical verification

Paper 38 records the explicit Λ values at small cutoffs:

$$\Lambda(\mathcal{T}_2, \mathcal{T}_{S^3}) \approx 2.0746, \quad \Lambda(\mathcal{T}_3, \mathcal{T}_{S^3}) \approx 1.6101, \quad \Lambda(\mathcal{T}_4, \mathcal{T}_{S^3}) \approx 1.3223, \tag{13}$$

with $\Lambda(4)/\Lambda(2) = 0.637$. These are monotone decreasing and consistent with the $O(\log n_{\max}/n_{\max})$ rate.

4.8 Tensor extension (Paper 39)

Paper 39 extends Theorem 4.1 to tensor products $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ of two copies of the same manifold at distinct focal lengths $\lambda_a \neq \lambda_b$ [35]. The result is a proved-at-qualitative-rate theorem closing the published-open NCG question of convergence for the tensor of two infinite metric spectral triples. Two technical points underwrite the closure:

- **R1: asymmetric Pythagorean Leibniz.** The Pythagorean Leibniz bound on tensor commutators (Connes–Marcolli graded form) combined with an asymmetric-supremum correction gives the full operator-system constant $C_3^{(2)}$ that is bounded away from $2\sqrt{2}$ and approaches the single-factor $C_3 = 1$ asymptotically.
- **R2: Latrémolière 2017 explicit $\varepsilon_{\text{cross}}$.** The cross term in the tunneling pair is bounded explicitly by Latrémolière [15] §4, with rate $O(\max \gamma)$ and constant $\leq 2\sqrt{2}$.

The bit-identical convergence ratio $\Lambda(n_{\max}, n_{\max})/\Lambda(2,2)$ on the diagonal matches the single-factor Paper 38 ratio numerically, providing a load-bearing sanity check.

5 The universality of the rate constant $4/\pi$

Paper 40 [36] extends Theorem 4.1 from SU(2) to the class of all simple compact Lie groups with bi-invariant metric.

Theorem 5.1 (Paper 40, universal rate constant)**.** *Let G be a simple compact connected Lie group of rank $r \geq 1$ with bi-invariant metric (so $G \in \{\mathrm{SU}(N), \mathrm{Spin}(n), \mathrm{Sp}(n), G_2, F_4, E_6, E_7, E_8\}$). The truncated Camporesi–Higuchi triples $\mathcal{T}_{n_{\max}}(G)$ converge to $\mathcal{T}_G$ in the Latrémolière propinquity with the same asymptotic rate constant as (6): namely $4/\pi$, in the dual-Coxeter normalisation. The constant $C_3(G) = 1$ remains asymptotically tight at all ranks.*

Plancherel-weight $\times$ Vandermonde-Jacobian cancellation. The proof of Theorem 5.1 runs through a structural identity which we record because it is the deepest algebraic content of the universality. In the central spectral Fejér kernel on G,

$$K_{n_{\max}}(g) \;=\; \frac{1}{Z_{n_{\max}}(G)} \left| \sum_{\lambda \,:\, |\lambda+\rho|^2 \leq M(n_{\max})^2} \sqrt{\dim V_\lambda}\, \chi_\lambda(g) \right|^2, \tag{14}$$

the Plancherel weight $\sqrt{\dim V_\lambda}$ on the squared amplitude (so the weight is $\dim V_\lambda$) and the Vandermonde Jacobian $|\Delta(t)|^2$ in the Weyl integration formula on G share the same $\prod_{\alpha>0}$ product structure over positive roots. In the dual-Coxeter normalisation these products cancel exactly, leaving only the rank-1 Stein–Weiss constant $4/\pi$ from each positive-root direction's Euler–Catalan asymptote $\sum_{d \text{odd}} 1/d^2 = \pi^2/8$. The universal constant $4/\pi$ is then *independent* of rank, Weyl group order, and Lie type.

Asymptotic tightness $C_3(G) = 1$ via PRV. The Lipschitz comparison bound (10) extends to G via the PRV-summand theorem (Kumar 1988, Vinberg 1990): for every Weyl group element $w \in W$, the dominant image σ_w of $\lambda + w\lambda'^*$ is a summand of $V_\lambda \otimes V_{\lambda'}^*$, and $|\sigma_w + \rho|^2 \geq |\lambda - \lambda'|^2 + |\rho|^2$. This covers the asymptotic family $(n_{\max}, 0, \ldots, 0) \otimes (1, 0, \ldots, 0)^*$ rigorously, giving $C_3(G) = 1$ asymptotic-tight in all simple Lie algebra types $A_n, B_n, C_n, D_n, G_2, F_4, E_6, E_7, E_8$. Interior (non-PRV) summands at finite cutoff are verified numerically across the panel $\mathrm{SU}(3), \mathrm{SU}(4), \mathrm{Sp}(2), G_2$ at 5641/5641 cells; analytic closure via Brauer–Klimyk signed-sum cancellation is bookkeeping.

Companion: state-space GH. The state-space Gromov–Hausdorff convergence (without rate and without Dirac) at the same generality was proved by Gaudillot-Estrada–van Suijlekom (IMRN 2025, paper rnaf197). Paper 40 provides the missing rate constant, the Lipschitz spinor bound, and the asymptotic tightness, lifting Gaudillot-Estrada–vS from state-space GH to spectral-triple-level propinquity convergence with explicit rate.

6 Modular structure and Wick rotation

6.1 The four-witness Wick-rotation theorem (Paper 42)

The Wick rotation between Euclidean and Lorentzian field theory is classically formulated through four equivalent presentations on a flat spacetime:

- Hartle–Hawking 1976 [10]: the temperature of the Euclidean cigar's τ-circumference identifies as the Hawking temperature;
- Sewell 1982 [26]: the KMS state at inverse-temperature $\beta = 2\pi/\kappa_g$ is the wedge vacuum;
- Bisognano–Wichmann 1976 [1]: the modular automorphism group of the wedge von Neumann algebra acts as boosts around the wedge edge;
- Unruh 1976 [29]: an accelerated observer with proper acceleration a sees a $T_U = a/(2\pi)$ thermal bath in the vacuum.

On a flat background, the four are equivalent under standard Wick-rotation arguments. On a curved background like S^3 they are not obviously equivalent, and at finite cutoff (operator-system level) they require separate construction.

Paper 42 [37] establishes the four-witness theorem at the operator-system level on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ via two structurally independent but operator-action conjugate constructions:

Theorem 6.1 (Paper 42, BW-α geometric realisation)**.** *For each $n_{\max} \in \{2, 3, 4, 5\}$, there exists a wedge projection $P_W = \frac{1}{2}(I + R_{\text{polar}})$ on $P_{n_{\max}}\mathcal{H}_{\text{GV}}$ (bit-exact idempotent and Hermitian) and a geometric modular Hamiltonian*

$$K_\alpha^W \;=\; J_{\text{polar}} \quad \textit{(rotation generator preserving } W\textit{)}, \tag{15}$$

with integer spectrum $\{\text{two}\, m_j\}$ on the full-Dirac basis. The modular flow satisfies $\sigma_{2\pi}(O) = O$ bit-exact (residual $O(\sqrt{\dim \mathcal{H}} \cdot \varepsilon_{\text{machine}})$) for the four witnesses BW, Hartle–Hawking, Sewell, Unruh, with cross-witness collapse residual identically zero.

Theorem 6.2 (Paper 42, BW-γ Tomita–Takesaki realisation)**.** *On the Krein-GNS Hilbert–Schmidt space $\mathcal{H}_{\text{GNS}} = M_{\dim W}(\mathbb{C})$ with KMS state $\rho_W = e^{-K_\alpha^W}/Z$ and the BW choice $H_{\text{local}} := K_\alpha^W/\beta$, the polar decomposition $S = J_{\text{TT}} \cdot \Delta^{1/2}$ gives a Tomita-Takesaki modular Hamiltonian $K_{\text{TT}} = -\log \Delta$ with the same integer spectrum, and the operator-action conjugacy*

$$\sigma_t^{\text{TT}}(a) \;=\; \sigma_{-t}^{\alpha}(a) \tag{16}$$

holds bit-exact at every tested $t \in \{1, \pi, 2\pi\}$. The modular operator Δ satisfies $J_{\text{TT}}^2 = +I$, categorically distinct from the Connes J_{GV} (KO-dim 3, $J^2 = -I$).

Load-bearing scope finding (Paper 42 §7.2). A key derived structural finding is that the local Hamiltonian required by Tomita–Takesaki at $\beta = 2\pi$ *must* be $H_{\text{local}} = K_\alpha^W/\beta$ rather than the intrinsic Camporesi–Higuchi Dirac operator D_{CH}. The intrinsic D_{CH} has half-integer spectrum, so $\beta \cdot D_{\text{CH}}$ would not have integer eigenvalues and the bit-exact closure $e^{i \cdot 2\pi \cdot n} = 1$ would fail. Spectral-action (Chamseddine–Connes [5]) and thermal-time (Connes–Rovelli) paradigms point to different generators on the wedge KMS state. This is the open question O3 of Paper 42 §10, sharpened in Paper 43 to the signature-independence statement below.

6.2 Lorentzian extension at finite cutoff (Paper 43)

Paper 43 [38] extends the four-witness theorem to the Lorentzian Krein space at signature $(3, 1)$, built via the Bizi–Brouder–Besnard prescription [2] at $(m, n) = (4, 6)$ in the Peskin–Schroeder chiral basis, with Lorentzian Dirac

$$D_L \;=\; i\left(\gamma^0 \otimes \partial_t \;+\; D_{\text{GV}} \otimes I_{N_t}\right) \tag{17}$$

per van den Dungen 2016 Proposition 4.1 [30].

Theorem 6.3 (Paper 43 §5–§7, Lorentzian four-witness at finite cutoff)**.** *At every tested* $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$*, the Krein-level analogues of* (15) *and* (16) *hold bit-exact:*

- *(i) The geometric Lorentzian modular Hamiltonian* $K_L^{\alpha,W} = J_{\text{polar}}|_{W_L}$ *has integer spectrum on the Krein hemispheric wedge.*
- *(ii) The Krein-Tomita modular operator* Δ_L *admits a polar decomposition* $S = J_{L,\text{TT}} \cdot \Delta_L^{1/2}$*; the Tomita modular Hamiltonian* K_L^{TT} *has integer spectrum.*
- *(iii)* $\sigma_{2\pi}^{L,\alpha}(O) = O$ *and* $\sigma_{2\pi}^{L,\text{TT}}(a) = a$ *bit-exact, with cross-witness collapse residual zero.*
- *(iv) At* $N_t = 1$*, the Lorentzian construction reduces to the Riemannian one of Theorems 6.1–6.2 bit-identically.*

Signature-independence of the load-bearing scope. The deepest structural finding of Paper 43 is that the "$H_{\text{local}} \neq D_W^L$" statement is *signature-independent* at the Riemannian reduction. At $N_t = 1$ the residual norm $\|H_{\text{local}} - D_W^L\|_F$ on the Lorentzian Krein wedge equals the Riemannian-side value at every $n_{\max}$ (Paper 43 §7.1):

$$\|H_{\text{local}}|_{(3,1),N_t=1} - D_L^W|_{N_t=1}\|_F \;=\; \|H_{\text{local}}^{\text{Riem}} - D_W\|_F \qquad (\text{at every } n_{\max}).$$

At $N_t > 1$ the Lorentzian residual is refined upward by the temporal-derivative content of D_L: the spectral-action object D_L has more content than D_{GV}, so the gap between H_{local} (modular-Hamiltonian generator) and D_W^L (spectral-action object) widens with temporal cutoff. This refines the Paper 42 §10 open question O3 from "open about signature-dependence" to a deeper structural feature.

6.3 Pythagorean HS-orthogonality and the $1/\pi^2$ signature

Paper 43 §10.2 records a Pythagorean refinement of the modular vs spectral-action gap. Let $\langle \cdot, \cdot \rangle_{\text{HS}}$ denote the Hilbert–Schmidt inner product on the wedge operator algebra.

Corollary 6.4 (Paper 43 §10.2, HS-orthogonality on the Lorentzian wedge)**.** *For every* $\kappa_g > 0$ *and every panel cell* $n_{\max} \in \{1, \ldots, 6\}$*,* $N_t \in \{1, 11\}$*, and witness* $w \in \{\text{BW}, \text{HH}_1, \text{HH}_2, \text{Sew}, \text{Unruh}_1, \text{Unruh}_2\}$*:*

$$\langle H_{\text{local}}, D_W^L \rangle_{\text{HS}} \;=\; 0 \qquad (\textit{bit-exact}), \tag{18}$$

and the residual norm has the closed form

$$\|H_{\text{local}} - D_W^L\|_F^2 \;=\; \frac{\kappa_g^2}{4\pi^2}\, S(n) + D(n), \quad S(n) = \frac{n(n+1)(n+2)(2n^2+4n-1)}{15}, \quad D(n) = \frac{n(n+1)(n+2)(2n}{20} \tag{19}$$

The closed form (19) is PSLQ-verified at 100 digits with coefficient ceiling 10^6 at 18 panel cells. The $1/\pi^2$ prefactor on $\|H_{\rm local}\|^2$ is the same M_1 Hopf-base measure signature (9) that appears in the Paper 38 GH convergence rate constant $4/\pi$. This crosswiring between sections 4 and 6 is taken up systematically in §8.

Subspace decomposition. The mechanism is the orthogonal decomposition of the operator algebra $B(\mathcal{K}_W)$ into a diagonal subspace (where $H_{\rm local} = K_\alpha^W/\beta$ lives, since K_α^W has eigenvalues two m_j on the full-Dirac basis) and an off-diagonal subspace with $\Delta n = \pm 1$ (where D_W^L lives, since the Camporesi–Higuchi Dirac intertwines $\pm m_j$ chirality partners at adjacent shells). Formal operator-theoretic proof of this decomposition is named as Paper 43 open question O4; the PSLQ-verified Pythagorean closed form (19) is empirically sharp at 100 digits and 18 panel cells.

7 Lorentzian propinquity

7.1 Operator-system substrate (Paper 44)

The first step in any Lorentzian propinquity proof is an operator- system substrate. Paper 44 [39] provides it on the chirality-doubled Camporesi–Higuchi spinor space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\rm GV}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ at BBB signature $(m, n) = (4, 6)$.

Theorem 7.1 (Paper 44, propagation number and envelope dependence)**.** *The truncated Lorentzian operator system $\mathcal{O}^L_{n_{\max},N_t}$ satisfies*

$$\mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) \;=\; 2 \quad \textit{under the Weyl-doubled achievable envelope}, \quad \mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) \;=\; \infty \quad \textit{under the full (di} \tag{20}$$

generalising the Riemannian prop $= 2$ *result for $\mathcal{O}_{n_{\max}}$ (Paper 32 §3.4) to the Lorentzian Krein setting. The witness pair $(M^{2,1,0,0}, M^{2,1,0,0*})$ at $n_{\max} = 2$ exhibits $\sim 38\%$ Frobenius residual against $\mathcal{O}^L_{2,3}$. Riemannian-limit recovery at $N_t = 1$ is bit-exact (Frobenius residual $= 0.0$ in float64) across $n_{\max} \in \{1, 2, 3\}$.*

K^+-positivity is trivial at the operator-multiplier level. A key structural observation of Paper 44 is that all chirality- doubled scalar multipliers $M \oplus M$ commute with $J_L =$ chirality-swap, and therefore preserve $\mathcal{K}^+$ automatically: $\mathcal{O}^{L,+} = \mathcal{O}^L$ exactly. Consequently the non-trivial Krein-positivity program shifts from operator level to *state* level (Wasserstein–Kantorovich on K^+ states), which is the natural setting for the Paper 45 propinquity proof below.

7.2 K^+-restricted weak-form Lorentzian propinquity convergence (Paper 45)

Paper 45 [40] closes the convergence theorem on the substrate of Paper 44.

Definition 7.2 (Paper 45 §1.3, K$^+$-weak-form Lorentzian propinquity)**.** For two truncated Lorentzian spectral triples $\mathcal{T}_i^L$ $(i = 1, 2)$, define

$$\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) \;:=\; \Lambda(\mathcal{T}_1^L|_{\mathcal{K}^+}, \mathcal{T}_2^L|_{\mathcal{K}^+}), \tag{21}$$

where Λ is the standard Latrémolière propinquity [15, 16] and the restriction to $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$ is the standard metric spectral triple obtained from the Krein-positive subspace, on which the indefinite Krein product reduces to a positive-definite Hilbert inner product.

Theorem 7.3 (Paper 45, Theorem 5.1)**.** *For the truncated* $\mathrm{SU}(2) \times \mathrm{U}(1)_T$ *Krein spectral triples $\mathcal{T}^L_{n_{\max},N_t,T}$, with D_L as in (17) and $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\rm GV}^{n_{\max}} \otimes \mathbb{C}^{N_t}$,*

$$\Lambda^L(\mathcal{T}^L_{n_{\max},N_t,T}, \mathcal{T}^L_{\mathcal{M}}) \;\le\; C_3^{\rm joint} \cdot \gamma^{\rm joint}_{n_{\max},N_t,T} \;\longrightarrow\; 0 \quad \textit{as } (n_{\max}, N_t) \to (\infty, \infty), \tag{22}$$

with $C_3^{\text{joint}} \leq 1$ *asymptotically tight (inherited from Paper 38 L3 via the joint Lichnerowicz structural identity) and*

$$\gamma^{\text{joint}}_{n_{\max},N_t,T} \;=\; O\big(\log n_{\max}/n_{\max} \;+\; T/N_t\big). \tag{23}$$

7.3 Three structural simplifications

The proof of Theorem 7.3 transfers the Paper 38 five-lemma chain to the joint $\mathrm{SU}(2) \times \mathrm{U}(1)$ setting. Three structural simplifications arise that are not visible in the Riemannian setting, and are recorded as substantive new content of the Paper 45 proof:

PURE_TENSOR Berezin map. The joint Berezin map $B^{\text{joint}} = B^{\mathrm{SU}(2)} \otimes B^{\mathrm{U}(1)}$ factorises as a pure tensor. Each of the four required properties (positivity, contractivity, approximate identity, L3 compatibility) inherits factor-by-factor; no cross-factor verification is needed.

Vanishing time-chirality cross-term. The joint Lichnerowicz commutator simplifies via the structural identity

$$[D_L,\, a_s \otimes a_t] \;=\; i\,[D_{\mathrm{GV}}, a_s] \otimes a_t \tag{24}$$

because $\{\gamma^0, D_{\mathrm{GV}}\} = 0$ on the chirality-doubled basis. (γ^0 anticommutes with the spatial Camporesi–Higuchi Dirac because D_{GV} is chirality-diagonal in the Peskin–Schroeder chiral basis.) The temporal direction contributes nothing to the joint commutator, and $C_3^{\text{joint}} = C_3^{\mathrm{SU}(2)}$ verbatim.

N_t-independent joint cb-norm. By the Bożejko–Fendler central-multiplier equality [3] on the amenable group product $\mathrm{SU}(2) \times \mathrm{U}(1)$, the joint cb-norm of the central spectral Fejér multiplier is

$$\|S_{K^{\text{joint}}}\|_{\mathrm{cb}} \;=\; \frac{2}{n_{\max}+1}, \tag{25}$$

independent of N_t. This reflects the trivial central-multiplier structure of the compact abelian factor $\mathrm{U}(1)$, and means that the N_t-dependence in (23) lives entirely in the joint mass-concentration moment (the T/N_t Fejér-on-the-circle rate), not in the cb-norm height bound.

Riemannian-limit recovery at $N_t = 1$ bit-exact. At $N_t = 1$, the joint Berezin reduces bit-identically to $B^{\mathrm{SU}(2)} \otimes B^{\mathrm{U}(1)}_{1,T} = B^{\mathrm{SU}(2)}$, since the $N_t = 1$ temporal Berezin map is the identity on the single Fourier mode. Hence $\Lambda^L(\mathcal{T}^L_{n_{\max},1,T}, \mathcal{T}^L_{\mathcal{M}})$ agrees with the Riemannian Paper 38 bound bit-exactly, providing the load-bearing falsifier check for the joint construction.

7.4 Honest scope

Theorem 7.3 is, to the author's knowledge, the first published Lorentzian propinquity convergence theorem in the math.OA literature. Several aspects of its scope require explicit comment.

(Q1) **Strong-form Lorentzian propinquity (Paper 45 §1.4 G1).** The K^+-restriction reduces the indefinite Krein product to a Hilbert inner product on $\mathcal{K}^+$ and lets the standard Latrémolière machinery apply verbatim. Strong-form Lorentzian propinquity (Latrémolière-style metric on Krein spectral triples in their own right, without K^+-restriction) requires an indefinite-inner-product analog of the operator-norm Lipschitz seminorm with appropriate behaviour under Krein-self-adjoint D_L. This remains an open NCG-math problem; multi-month scale.

(G2) **De-compactification $T \to \infty$.** The temporal factor is compactified to $\mathrm{U}(1)_T$ with finite radius T. De-compactification to non-compact $\mathbb{R}_t$ is a separate program (internal Sprint L3c) requiring renormalisation of the cb-norm bound and reach moment.

(G3) **Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$.** Blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry recorded in Paper 24 §V of the GeoVac corpus (a four-layer Coulomb/HO asymmetry); this is the Coulomb-side claim only.

(G4) **Inner-factor calibration data.** Higgs / Yukawa selection question of the Connes–Marcolli SM construction; orthogonal to the present convergence theorem, addressed by Paper 32 §8.C verdict POSITIVE-THIN (construction admits Higgs structurally, but no autonomous Yukawa selection from the present data).

8 The master Mellin engine and the cross-arc connection

The most striking structural observation that emerges from reading the arc is that the same transcendental signatures appear in *different* settings. The constant $4/\pi$ of Paper 38 and the constant $1/\pi^2$ of Paper 43 §10.2 are not numerical coincidences: they are two appearances of the same Hopf-base measure mechanism, the M_1 sub-case of the master Mellin engine catalogued in Paper 32 §8.

8.1 The π-source case-exhaustion theorem (Paper 32 §8)

Theorem 8.1 (Paper 32 §8, π-source case-exhaustion). *Let $\mathcal{C}$ be any finite composition of GeoVac projections applied to a Layer-1 datum on $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ or its Bargmann–Segal sibling triple. The following are equivalent:*

(i) $\mathcal{C}$(datum) contains a transcendental of the form π^k for $k \in \mathbb{Q}$;

(ii) $\mathcal{C}$ contains at least one projection whose evaluation includes a continuous integration over a parameter promoted from the discrete graph spectrum;

(iii) $\mathcal{C}$ engages at least one of three spectral-triple mechanisms,

- M_1 *(*Hopf-base measure*): $\pi = \mathrm{Vol}(S^2)/4$ via the Hopf bundle $S^3 \to S^2 \times S^1$, equivalently $4/\pi = \mathrm{Vol}(S^2)/\pi^2$;*
- M_2 *(*Seeley–DeWitt*): $\sqrt{\pi}$ via heat-kernel coefficients on the Camporesi–Higuchi Dirac spectrum on S^3 (Paper 28 Theorem T9: $\zeta_{D^2}(s)$ is polynomial in π^2 with rational coefficients at every integer s);*
- M_3 *(*vertex-topology Dirichlet L*): Catalan G and $\beta(s)$ via quarter-integer Hurwitz shifts in two-loop sunset vertices.*

The TS-E3 sharpening of Paper 32 further shows that M_1, M_2, M_3 are sub-cases of a single master Mellin-engine mechanism

$$\pi\text{-source} \;=\; \mathcal{M}\big[\mathrm{Tr}(D^k e^{-tD^2})\big] \quad \text{with } k \in \{0, 1, 2\}, \tag{26}$$

where $k = 0$ selects M_1, $k = 1$ selects M_3, and $k = 2$ selects M_2.

8.2 M_1 in the Riemannian GH rate

The rate constant $4/\pi$ in (6) is the M_1 Hopf-base measure signature. This is not a derived observation; it is forced by the Plancherel-weight $\times$ Vandermonde-Jacobian cancellation theorem

(Paper 40 §3.2) which leaves only the rank-1 Stein–Weiss constant after the $\prod_{\alpha>0}$ products cancel. Equivalently, the single-positive-root Euler–Catalan sum

$$\sum_{d \text{ odd}} \frac{1}{d^2} = \frac{\pi^2}{8}$$

produces the π^2 in the denominator, and the Hopf bundle $S^3 \to S^2$ produces $\mathrm{Vol}(S^2) = 4\pi$ in the numerator.

8.3 M_1 in the Lorentzian HS-orthogonality

The $1/\pi^2$ prefactor on $\|H_{\mathrm{local}}\|^2$ in the Pythagorean closed form (19) is the same M_1 signature *at the operator-system level*: the Krein wedge BW choice $H_{\mathrm{local}} = K_\alpha^W/(2\pi)$ introduces a factor of $1/(2\pi)$ from the modular-circle circumference, which squared gives $1/(4\pi^2)$ in $\|H_{\mathrm{local}}\|^2$. The Hopf-base measure factor $4/\pi$ of Paper 38 and the $1/\pi^2$ of Paper 43 are two manifestations of the same $\mathrm{Vol}(S^2) = 4\pi$ identity in the master Mellin engine.

Why this matters. The case-exhaustion theorem (Theorem 8.1) is a *spectral-triple-level* statement about which transcendentals can appear in evaluations of GeoVac projection chains. The constant $4/\pi$ of Paper 38 is a *metric-geometry-level* statement about the propinquity rate. The fact that these two abstractly-different levels agree on the same signature is non-trivial: it identifies the metric geometry of the GeoVac spectral triple as carrying the same M_1 signature that the abstract Mellin engine catalogues. Abstract taxonomy meets concrete propinquity rates. Paper 43 §10.2 takes this further: the Lorentzian Pythagorean decomposition at the operator-system level carries the same M_1 signature again, this time on the operator algebra of the Krein wedge.

9 Closures since drafting (May 23, 2026)

The synthesis above (§1–§8) was drafted around the arc Papers 29, 32, 38, 39, 40, 42, 43, 44, 45. On 2026-05-23, four substantive closures landed in a single session, adding Papers 46 and 47 to the arc and promoting two named open follow-ons within Papers 39 and 43. This section records the four closures in summary form.

9.1 Strong-form Lorentzian propinquity (Paper 46)

Paper 45 §1.4 G1 named the strong-form Lorentzian propinquity (a Latrémolière-style metric on Krein spectral triples without K^+-restriction) as the principal open NCG-math problem of the Lorentzian arc. Paper 46 [41] closes this question on the natural chirality-doubled scalar-multiplier substrate of Paper 44 [39] via the operator-norm Lipschitz seminorm $L_{\mathrm{op}}(a) := \|[D_L, a]\|_{\mathrm{op}}$:

$$\Lambda^{\mathrm{strong}}(\mathcal{T}^L_{n_{\max},N_t,T},\ \mathcal{T}^L_{\mathcal{M}}) \ \le\ C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \ \to\ 0, \tag{27}$$

with closed-form $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}} \to 1^-$ and rate $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ surviving verbatim from Paper 45. The numerical panel matches Paper 45's K^+-weak-form panel *bit-exactly* at every tested cell — the "free-upgrade" reading.

Paper 46's Appendix B (added 2026-05-22) extends the closure to the *enlarged* operator system with chirality-flipping generators satisfying $\{J_L, M^{\mathrm{flip}}\} = 0$. The substantive new content is the *gradient-norm absorption mechanism*: the strict-strong-form separation between enlarged and K^+-weak-form substrates manifests at the Lipschitz seminorm level ($L_{\mathrm{op}}(a^{\mathrm{flip}}) > 0$ on enlarged generators while $L^+_{\mathrm{P45}}(a^{\mathrm{flip}}) = 0$ identically), but the separation is paid *entirely in a J-graded gradient-norm extension* $G^{\mathrm{enlarged}} = G^{\mathrm{natural}} + 2\|D_t\|_{\mathrm{op}}\|f^{\mathrm{flip}}\|_{\mathrm{op}}$, leaving the Latrémolière propinquity bound bit-exactly preserved: $\Lambda^{\text{enlarged}}(n_{\max}, N_t) = \Lambda^{\text{P46}}(n_{\max}, N_t)$ at every panel cell.

9.2 De-compactification limit, norm-resolvent / spectral closure (Paper 47)

Paper 45 §1.4 G2 named the de-compactification limit $T \to \infty$ on the temporal carrier S^1_T as a separate open program. Paper 47 [42] closes it at the *norm-resolvent / spectral* level via a *two-rate hybrid composite*:

$$\mathcal{T}^L_{n_{\max},N_t,T(N_t)} \xrightarrow{\Lambda^L\text{-rate } O(\log n_{\max}/n_{\max}+T(N_t)/N_t)} \mathcal{T}^L_{S^3\times S^1_{T(N_t)}} \xrightarrow{\text{nr-rate } O(e^{-|\Im z|T(N_t)/2})} \mathcal{T}^L_{S^3\times\mathbb{R}_t} \tag{28}$$

along any *admissible scaling* $T(N_t) \to \infty$ with $T(N_t)/N_t \to 0$. The inner arrow is propinquity convergence at coupled cells (Paper 45/46 main theorem applied along the scaling); the outer arrow is norm-resolvent convergence of the Lorentzian Dirac on the compactified spacetime to the Lorentzian Dirac on the genuine spacetime, with explicit exponential tail bound $O(e^{-|\Im z|T/2})$ on compact-support subspaces via Reed–Simon [43] § XIII.16 / Bloch-decomposition machinery and the Krein-isometric embedding $\widetilde{J}_T : \; L^2(S^1_T) \hookrightarrow L^2(\mathbb{R})$.

A strategic clarification appears in Paper 47 §1.1: the natural *inductive-limit propinquity* framework of Latrémolière [16] § 5 for AF C*-algebras is structurally wrong for the present setting because AF C*-algebras have totally-disconnected Gelfand spectrum while $\mathbb{R}$ is connected, so $C_0(\mathbb{R})$ cannot be an AF inductive limit. Norm-resolvent convergence is the right framework at the spectral level.

Paper 47 §6 also establishes a *three-carrier identification*: the periodic compact $S^3 \times S^1_T$ (Paper 45 substrate), the bounded Dirichlet $S^3 \times [-T/2, T/2]$ (Paper 43 / Sprint L2-B/E substrate), and the non-compact $S^3 \times \mathbb{R}_t$ (Paper 47 limit) all converge in norm-resolvent sense to the same physical Lorentzian Dirac as $T \to \infty$. This resolves the apparent tension between Paper 43's bounded-interval architecture (chosen to preserve causal structure / no CTCs via Geroch's theorem) and Paper 45's periodic-compact architecture (chosen to enable Latrémolière propinquity machinery on a compact carrier). Both are valid compact approximations to the same physical Lorentzian Dirac.

A numerical-verification panel (Sprint L3c-α.2, 2026-05-23) at 24 cells $(n_{\max}, N_t) \in \{2,3\} \times \{3,5,7,11\}$ along three scalings ($T_0 = 2\pi$ canonical, $T_1 = N_t/\log N_t$ sub-linear, $T_2 = \sqrt{N_t}$ square-root) confirms Λ^L bit-exact T-and-N_t-independence at fixed $n_{\max}$: $\Lambda(n_{\max} = 2) = 2.074551$ and $\Lambda(n_{\max} = 3) = 1.610060$ to 6+ significant figures across all three scalings, matching Paper 45 Table 1 verbatim. Structural reason: $\gamma^{U(1)}(N_t, T)$ is subordinate to $\gamma^{\text{SU}(2)}(n_{\max})$ at every admissible-scaling cell, so the joint rate $\gamma^{\text{joint}} = \max(\gamma^{\text{SU}(2)}, \gamma^{U(1)}) = \gamma^{\text{SU}(2)}$ exactly. The empirical confirmation upgrades the L3c-α theorem from "analytical corollary of Paper 45" to "analytically AND empirically established."

9.3 Pythagorean orthogonality formal proof (Paper 43 §10.2 promotion)

Paper 43 §11 O4 named a formal proof of the diagonal/off-diagonal subspace-decomposition mechanism behind the L2-F.1 (2026-05-17) empirical observation $\langle H_{\text{local}}, D^W_L\rangle_{\text{HS}} = 0$ as a follow-on. The formal proof landed on 2026-05-23 via a $\mathbb{Z}_2$ wedge-chirality parity argument and is now recorded in Paper 43 §10.2 as a formal theorem.

The argument has three structural inputs from Papers 42/43/44 [37–39]: (I1) the wedge chirality grading Π_W (BBB Krein-chirality γ^5 restricted to the wedge) is a unitary self-adjoint involution; (I2) $[\Pi_W, H_{\text{local}}] = 0$ (even parity of the geometric BW-α generator $K^W_\alpha = J^W_{\text{polar}}$ under wedge chirality); (I3) $\{\Pi_W, D^W_L\} = 0$ (odd parity of the wedge-restricted Lorentzian Dirac, via $\{\gamma^0, \gamma^5\} = 0$ on the temporal factor and the unfolded-wedge basis change on the spatial factor).

The orthogonality follows from a parity / cyclicity-of-trace lemma: for any unitary involution Π on a Hilbert space, $[\Pi, A] = 0$, $\{\Pi, B\} = 0$ imply $\mathrm{Tr}(A^\dagger B) = 0$ via $\mathrm{Tr}(X) = \mathrm{Tr}(\Pi X \Pi^{-1})$ giving $\mathrm{Tr}(A^\dagger B) = \mathrm{Tr}(A^\dagger \cdot (-B)) = -\mathrm{Tr}(A^\dagger B)$, forcing 0. The bit-exact L2-F.1 numerical panel at 18 cells is empirical corroboration of the now-algebraic identity at machine precision.

9.4 Master theorem of the math.OA arc

Paper 39 §6 (iii) and (v) named two open follow-ons: (iii) the k-fold extension of the Paper 39 two-factor tensor-product propinquity to $\mathcal{T}_{S^3}^{\otimes k}$, and (v) the higher-rank tensor product $\mathcal{T}_{G_1} \otimes \mathcal{T}_{G_2}$ at arbitrary compact connected Lie group factors. Both closed on 2026-05-23.

(iii) k-fold extension — k-independent constant. The sprint memo `debug/sprint_paper39_k_fold_e` established the k-fold theorem at SU(2) factors with closed-form $C_3^{(k)} \le \sqrt{(N_* - 1)/(N_* + 1)}$ where $N_* := \max_j n_j$ — **k-INDEPENDENT, identical to the single-factor Paper 38 L3 constant**, asymptotic $\to 1^-$. The $O(\sqrt{k})$ estimate that the GeoVac internal record had carried (triangle-inequality bound) was strictly improved by the k-fold Pythagorean Leibniz identity $\|\sum_j X_j\|_{\mathrm{op}}^2 = \sum_j \|X_j\|_{\mathrm{op}}^2$ on the Connes-Marcolli graded module — the k summands of $D^{(k)}$ pairwise anti-commute, so the cross-terms vanish and the sum-of-magnitudes bound is replaced by the sum-of-squares identity. The improvement of $\sqrt{k}$ comes from precisely the Pythagorean refinement of the triangle inequality.

(v) Master theorem — universal compact-Lie-group factors. Combining the k-fold extension with the universal $4/\pi$ rate of Paper 40 [36] produces the master theorem of the math.OA arc:

$$\boxed{\Lambda^{(k)}\big(\mathcal{T}^{(k)}_{n_1,\ldots,n_k},\ \mathcal{T}^{(k)}_{G_1\times\cdots\times G_k}\big) \ \le\ \sup_j C_3^{G_j}(n_j) \cdot \max_j \gamma_{n_j}^{G_j}} \tag{29}$$

for arbitrary $k \ge 1$ and arbitrary compact connected Lie groups $G_1, \ldots, G_k$ with bi-invariant metrics. $C_3^{(k)}$ is k-INDEPENDENT (bounded by the worst-factor Paper 40 L3 constant, $\to 1^-$). The joint rate inherits the universal $4/\pi$ asymptotic from Paper 40's Plancherel-weight $\times$ Vandermonde-Jacobian cancellation.

The two extensions are *structurally orthogonal*: the k-fold mechanism (Pythagorean Leibniz) acts on the factor-counting dimension; the general-G mechanism (PRV-summand Weyl harmonics) acts on the per-factor structure. Each preserves the other's content cleanly.

The master theorem subsumes Papers [34] ($k = 1$, $G = \mathrm{SU}(2)$), [36] ($k = 1$, general G), and the two-factor case treated above (§4.8, $k = 2$, $G_1 = G_2 = \mathrm{SU}(2)$) as special cases. Paper 45's compact-temporal Lorentzian extension at $S^3 \times S^1_T$ is the $k = 2$, $G_1 = \mathrm{SU}(2)$, $G_2 = \mathrm{U}(1)$ special case, providing independent justification of Paper 45's main theorem from the master construction.

10 Open questions and outlook

The arc closes the convergence side of the operator-algebra story at the K$^+$-weak-form level (Paper 45) and the modular-Hamiltonian side at the operator-system level on Riemannian (Paper 42) and Lorentzian (Paper 43) backgrounds. Several questions remain open.

Strong-form Lorentzian propinquity (Q1) — CLOSED by Paper 46 (§9.1). As noted in §7, the K$^+$-restriction is the move that lets the standard Latrémolière machinery apply verbatim. Strong-form Lorentzian propinquity, formulated as a Latrémolière-style metric on Krein spectral triples in their own right (Paper 45 §1.4 G1), is now closed by Paper 46 [41] via the operator-norm Lipschitz seminorm $L_{\mathrm{op}}(a) = \|[D_L, a]\|_{\mathrm{op}}$ on the natural chirality-doubled scalar-multiplier substrate of Paper 44. The "free-upgrade" reading holds: $\Lambda^{\mathrm{strong}}$ matches the K^+-weak-form bound bit-exactly at every panel cell, with $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}} \to 1^-$.

Paper 46 Appendix B extends the closure to the enlarged operator system with chirality-flipping generators via a gradient-norm absorption mechanism, leaving the propinquity bound bit-exactly preserved.

De-compactification $T \to \infty$ (G2) — CLOSED at norm-resolvent level by Paper 47 (§9.2); G2-metric open as multi-month frontier. The temporal factor $\mathrm{U}(1)_T$ is finite; the standard physics applications (Hawking, Unruh) require non-compact $\mathbb{R}_t$. The rate $T/N_t \to 0$ in (23) is the joint mass-concentration on $\mathrm{U}(1)_T$ as $N_t \to \infty$ at fixed T; the $T \to \infty$ limit is closed at the *norm-resolvent / spectral* level by Paper 47 [42]'s two-rate hybrid composite (28). The remaining open question is the *metric / propinquity* level closure on a non-compact carrier (G2-metric or equivalently G1-genuine on the non-compact substrate), which requires a non-compact extension of Latrémolière's propinquity not in the published literature as of May 2026. This is named multi-month frontier (Sprint L3e in our internal taxonomy), with the path-of-least-resistance being a pointed-propinquity bridge to the Mondino–Sämann synthetic Lorentzian GH program (§9.2).

Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ (G3, Coulomb-side claim only). The natural tensor product of the Coulomb S^3 spectral triple with the Bargmann–Segal Hardy-sector spectral triple on S^5 [31] is blocked at the NCG-framework level by a four-layer asymmetry recorded in Paper 24 §V. The four layers are: (i) spectrum-computing role of L_0 (Coulomb-side only, by Paper 23 Fock rigidity); (ii) calibration π (Coulomb-side only); (iii) non-abelian Wilson gauge with natural matter coupling (both sides have Wilson, only Coulomb-side has natural matter coupling because the Bargmann tower $(N, 0)$ irreps have N-dependent dimensions and no uniform-rep matter sector); and (iv) modular- Hamiltonian structure of the wedge KMS state (Coulomb-side admits the Krein wedge HS-orthogonality of Corollary 6.4 with closed-form M_1 prefactor; the Hardy-sector side does not admit the construction at all). Closing G3 requires new NCG-framework machinery to bridge Riemannian and Hardy-sector spectral triples in a single tensor product; we make no claim about whether this is achievable.

Higgs as inner fluctuation (G4). The GeoVac triple is naturally combined with a finite almost- commutative factor $\mathcal{A}_{\mathrm{GV}} \otimes (\mathbb{C} \oplus \mathbb{H})$ to produce an electroweak-like sector. The Connes axiom audit at finite cutoff (Paper 32 §8.C) gives verdict POSITIVE-THIN: the construction *admits* a Higgs sector structurally (i.e. the inner fluctuation $D \mapsto D + \omega + \varepsilon' J \omega J^{-1}$ produces a non-trivial off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ block iff the finite Yukawa Y_F is non-zero), but *GeoVac does not autonomously select* Y_F from its internal data. In Marcolli–vS terms, GeoVac sits on the Yang–Mills-without-Higgs side of the Perez-Sanchez [22, 23] clarification, with calibration data (the Yukawa) supplied externally. Paper 32 §8.C records this as the standing structural verdict.

State-side complement of the dictionary. Paper 44 highlights that the non-trivial K^+-positivity content of the Lorentzian operator system sits at the *state* level, not the operator level. The natural object is the Wasserstein–Kantorovich metric on K^+-states. Paper 45's K^+-weak-form propinquity is the first concrete construction in this direction; substantially more mathematics on Wasserstein–Kantorovich on Krein-positive states is required to complete the state-side dictionary.

Concluding remark. The structural identification of GeoVac as an almost-commutative spectral triple in the Marcolli–van Suijlekom lineage is proved at the GH-convergence level by Paper 38 (Riemannian, single factor), extended by Paper 39 (tensor product), generalised by Paper 40 (all compact simple Lie groups), and extended in turn to the Lorentzian K^+-weak-form by Paper 45. The metric geometry of the truncated triples carries the same M_1 Hopf-base measure signature ($4/\pi$ in the Riemannian rate, $1/\pi^2$ in the Lorentzian Pythagorean closed form) that the abstract π-source case-exhaustion theorem (Theorem 8.1) identifies at the spectral-triple level. The cross-arc identification of these signatures is the deepest structural observation of the corpus.

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